QUEUEING PROCESS

$$\begin{tikzcd} 0 \arrow[r, bend left, "6"] & 1 \arrow[l, bend left, "3"] \arrow[r, bend left, "6"] & 2 \arrow[l, bend left, "6"] \arrow[r, bend left, "6"] & 3 \arrow[l, bend left, "9"] \arrow[r, bend left, "6"] & 4 \arrow[l, bend left, "9"] \arrow[r, bend left, "6"] & 5 \arrow[l, bend left, "9"] \end{tikzcd}$$

IN = OUT

$$\begin{cases} 3\pi_1 = 6\pi_0 \\ 6\pi_2 = 6\pi_1 \\ 9\pi_3 = 6\pi_2 \\ 9\pi_4 = 6\pi_3 \\ 9\pi_5 = 6\pi_4 \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 + \pi_5 = 1 \end{cases}$$

$$\pi_1 = 2\pi_0 \qquad \pi_2 = 2\pi_0 \qquad \pi_3 = \frac{4}{3}\pi_0 \qquad \pi_4 = \frac{8}{9}\pi_0 \qquad \pi_5 = \frac{48}{81}\pi_0$$

$$\pi_0 + 2\pi_0 + 2\pi_0 + \frac{4}{3}\pi_0 + \frac{8}{9}\pi_0 + \frac{48}{81}\pi_0 = 1$$

$$81\pi_0 + 162\pi_0 + 162\pi_0 + 108\pi_0 + 72\pi_0 + 48\pi_0 = 81$$

$$\pi_0 = \frac{81}{633} = \frac{27}{211}$$

$$\pi = \frac{1}{211}(27, 54, 54, 36, 24, 16)$$

NUMBER OF STUDENTS:

$$(0\cdot 27 + 1\cdot 54 + 2\cdot 54 + 3\cdot 36 + 4\cdot 24 + 5\cdot 16)\cdot\frac{1}{211} = \frac{446}{211} \approx 2.11$$

LET E_x = TIME to GO FROM STATE x to STATE 4

WE WANT to COMPUTE E_0 BECAUSE the EXAMINERS HAVE JUST STARTED and THERE ARE NO STUDENTS YET

TO FIND E_0 WE HAVE to SET UP A SYSTEM of EQUATIONS INVOLVING the WAITING TIMES in EACH STATE AND THE (TRANSITION) PROBABILITIES of JUMPING in ANOTHER STATE

- THE TRANSITION PROB CAN BE OBTAINED THANKS to the "LACK of MEMORY" PROPERTY of the EXPONENTIAL DISTRIBUTION. IN PARTICULAR, GIVEN $X \sim exp(\alpha)$, $Y \sim exp(\beta)$, THEN $\mathbb{P}(X < Y) = \frac{\alpha}{\alpha+\beta}$. IN THIS CASE, WHEN WE ARE IN STATE i, $X, Y \in \{i+1, i-1\}$ AND α and β ARE the RATES AS DESCRIBED IN THE GRAPH.

- THE WAITING TIMES RELATES to the EXPECTED VALUE of the EXPONENTIAL DIST DESCRIBING the ARRIVAL / SERVICE time

- IN PARTICULAR, GIVEN $X \sim exp(\alpha)$, THEN $\mathbb{E}[X] = \frac{1}{\alpha}$

WHEN IN STATE i, the EXPECTED TIME DEPENDS on the RATE AT WHICH WE LEAVE THE STATE EITHER jumping FORWARD OR jumping BACKWARD (α and β RESPECTIVELY). HENCE, SINCE GIVEN $X \sim exp(\alpha)$, $Y \sim exp(\beta)$, $Z = \min\{X, Y\}$ THEN $Z \sim exp(\alpha+\beta)$, THE RATE at WHICH WE LEAVE STATE i is $\alpha+\beta$ and the WAITING TIME is $\frac{1}{\alpha+\beta}$

STATE 0 is a PARTICULAR CASE, AS WE CAN LEAVE only to STATE 1 with RATE 6

STATE 4 is a PARTICULAR CASE, SINCE IT is the ONE WE WISH to REACH, HENCE $E_4 = 0$ BECAUSE the is NOTHING to WAIT.

ACCORDING to THIS DISCUSSION WE OBTAIN THE FOLLOWING SYSTEM

$$\begin{cases} E_0 = \frac{1}{6} + E_1 \\ E_1 = \frac{1}{9} + \frac{1}{3}E_0 + \frac{2}{3}E_2 \\ E_2 = \frac{1}{12} + \frac{1}{2}E_1 + \frac{1}{2}E_3 \\ E_3 = \frac{1}{15} + \frac{3}{5}E_2 + \frac{2}{5}E_4 \\ E_4 = 0 \end{cases}$$

$$\left[E_0 = \frac{39}{24} \sim 1,625 \text{ hours} \right]$$

ALTERNATIVE SOL:

LET E_x = TIME to GO FROM STATE x to STATE 4

WE WANT to COMPUTE E_0 BECAUSE the EXAMINERS HAVE just STARTED and there ARE NO STUDENTS yET

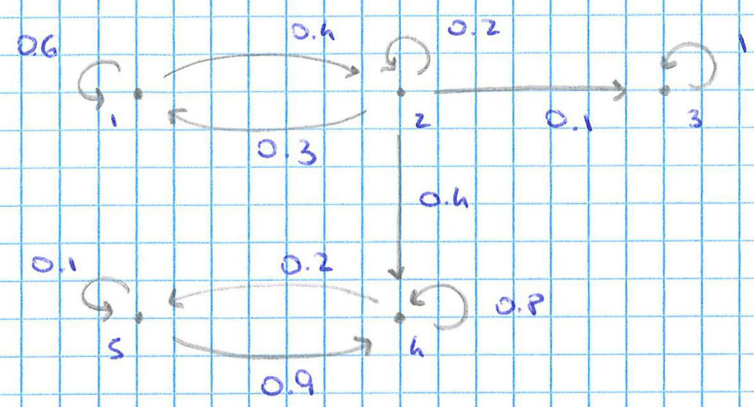
TO FIND E_0 WE HAVE TO SOLVE the FOLLOWING SYSTEM OF EQUATIONS

$$\begin{cases} E_0 = \frac{1}{6} + E_1 \\ E_1 = \frac{1}{9} + \frac{1}{3}E_0 + \frac{2}{3}E_2 \\ E_2 = \frac{1}{12} + \frac{1}{2}E_1 + \frac{1}{2}E_3 \\ E_3 = \frac{1}{15} + \frac{3}{5}E_2 + \frac{2}{5}E_4 \\ E_4 = 0 \end{cases}$$

- the TRANSITION PROB CAN BE OBTAINED...
- the WAITING TIMES ...

STATE 0 and STATE 4 ARE PARTICULAR CASES BECAUSE ...

MARKOV CHAINS



1. LET P_x be the PROBABILITY of ENDING in STATE 3 STARTING FROM X

$$\begin{cases} P_1 = \frac{6}{10} P_1 + \frac{4}{10} P_2 \\ P_2 = \frac{2}{10} P_2 + \frac{3}{10} P_1 + \frac{4}{10} P_4 + \frac{1}{10} P_3 \\ P_3 = 1 \\ P_4 = 0 \\ P_5 = 0 \end{cases}$$

$$P_2 = \frac{1}{5} P_2 + \frac{3}{10} P_1 + \frac{1}{10} \qquad P_2 = \frac{3}{8} P_1 + \frac{1}{8}$$

$$P_1 = \frac{3}{5} P_1 + \frac{2}{5}\left(\frac{3}{8} P_1 + \frac{1}{8}\right) \qquad P_1 = \frac{1}{5}$$ PROB OF ARRIVING in 3 STARTING in 1

$\Rightarrow \frac{4}{5}$ PROB OF ARRIVING in {4,5} STARTING in 1

2. A STATE IS a TRANSIENT STATE IF, UPON ENTERING this STATE, THE PROCESS MIGHT NEVER RETURN to THIS STATE AGAIN, i.e., WE VISIT SUCH STATE FINITELY OFTEN.

A STATE IS a RECURRENT STATE IF, UPON ENTERING THIS STATE, THE PROCESS WILL DEFINITELY RETURN to THIS STATE AGAIN, i.e., WE WILL VISIT THIS STATE INFINITELY OFTEN.

A STATE IS an ABSORBING STATE IF UPON ENTERING THIS STATE, the PROCESS WILL NEVER LEAVE THIS STATE AGAIN

THE PROVIDED MARKOV CHAIN CONTAINS ALL THESE TYPES OF STATES, IN PARTICULAR:

- {1,2} ARE TRANSIENT STATES
- {4,5} ARE RECURRENT STATES
- {3} IS ABSORBING

3. $P_T = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.2 \end{bmatrix}$

$$S = (I - P_T)^{-1} = \begin{bmatrix} 0.4 & -0.4 \\ -0.3 & 0.8 \end{bmatrix}^{-1}$$

$$= \frac{1}{0.4 \cdot 0.8 - 0.4 \cdot 0.3} \begin{bmatrix} 0.8 & 0.4 \\ 0.3 & 0.4 \end{bmatrix}$$

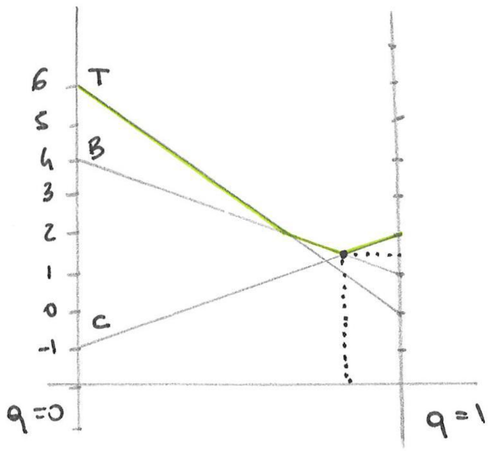
$$= \begin{bmatrix} 4 & 2 \\ 3/2 & 2 \end{bmatrix}$$

STARTING FROM 1, THE EXPECTED NUMBER OF STAGES IN STATE 1 IS 4
2 IS 2

GAME THEORY

$$\begin{array}{c} \\ A = \end{array} \begin{array}{c} \begin{bmatrix} 6 & 0 \\ -1 & 2 \\ 4 & 1 \end{bmatrix} \\ \begin{matrix} 1-q & q \end{matrix} \end{array} \begin{array}{c} p_1 \\ p_2 \\ p_3 \\ \\ \end{array}$$

No dominated rows or colums
No saddle points
=> mixed strategy



B $4(1-q)+q$

C $-(1-q)+2q$

$4(1-q)+q=-(1-q)+2q$

$4-3q=-1+3q$

$6q=5$

$q=\frac{5}{6}$ value $=\frac{3}{2}$ $\left(\frac{1}{6},\frac{5}{6}\right)$

$p_1=0$ TOP do not influence the strategy ~~of~~ P2

L $-p_2+4p_3$

R $2p_2+p_3$

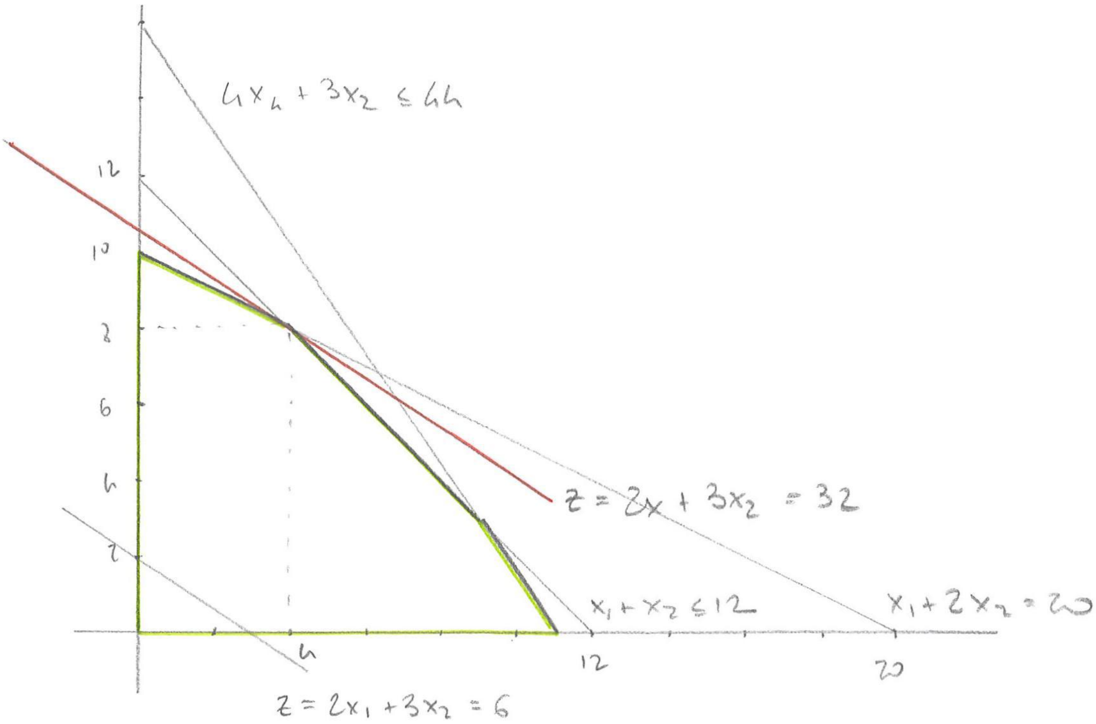
$-p_2+4p_3=2p_2+p_3$

$3p_3=3p_2$

$p_2=p_3$

$p_1+p_2+p_3=1$ $\left(0,\frac{1}{2},\frac{1}{2}\right)$

6.



1. DRAW THE FEASIBLE REGION (IN GREEN) ACCORDING TO THE CONSTRAINTS
2. DRAW THE LINE CORRESPONDING TO THE OBJECTIVE FUNCTION (E.G. STARTING FROM $z = 6$)
3. MOVE THE LINE UP (SINCE WE ARE MAXIMIZING) UNTIL WE REACH A CORNER OF THE FEASIBLE SET
4. THE POINT WE HAVE REACHED IS THE OPTIMAL SOLUTION

$x_1 = 4$
$x_2 = 8$
$z = 32$ (THIS VALUE CAN BE COMPUTED BY REPLACING x_1 and x_2 IN THE EQUATION for z)

SIMPLEX METHOD

$$\max \quad z = 2x_1 + 3x_2$$

st

$$x_1 + 2x_2 \le 20$$
$$x_1 + x_2 \le 12$$
$$4x_1 + 3x_2 \le 44$$
$$x_i \ge 0 \qquad i = 1,2$$

a. $\max \quad z$

st

$$z - 2x_1 - 3x_2 = 0$$
$$x_1 + 2x_2 + x_3 = 20$$
$$x_1 + x_2 + x_4 = 12$$
$$4x_1 + 3x_2 + x_5 = 44$$
$$x_i \ge 0 \qquad i = 1 - 5$$

ITER	BV	z	x_1	x_2	x_3	x_4	x_5	RHS	
0	z	1	-2	-3	0	0	0	0	
	x_3	0	1	2	1	0	0	20	$20/ \; = 10$
	x_4	0	1	1	0	1	0	12	$12/1 = 12$
	x_5	0	4	3	0	0	1	44	$44/3 > 11$
1	z	1	$-\frac{1}{2}$	0	$\frac{3}{2}$	0	0	30	
	x_2	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	10	$10/\frac{1}{2} = 20$
	x_4	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	0	2	$2/\frac{1}{2} = 4$
	x_5	0	$\frac{5}{2}$	0	$-\frac{3}{2}$	0	1	14	$14/\frac{5}{2} > 5$
2	z	1	0	0	1	1	0	32	
	x_2	0	0	1	1	-1	0	8	
	x_1	0	1	0	-1	2	0	4	
	x_5	0	0	0	1	-5	1	4	

$x_1 = 4 \qquad x_2 = 8 \qquad x_5 = 4$

$x_3 = 0 \qquad x_4 = 0$

$z = 32$

TRANSPORTATION PROBLEM

	1	2	3	supply	u_i
1	3 / 6	10 / − 6	2 / −13 + (E)	12	0
2	5 / 8	4 / + 9	9 / 3 − (L)	12	−6
3	5 / 7	10 / 5	10 / 6	6	−5
demand	6	15	9	Z = 201	
v_j	3	10	−15		

1. Basic feasible sol with North-West Corner rule with value Z
2. Compute u_i and v_j to initiate the next iteration and compute $c_{ij} - u_i - v_j$ for non-basic variables
3. Select most negative and check chain of motion to find entering E and leaving L basic variable

	1	2	3	s	u_i
1	3 / 6	10 / − 3 (L)	2 / 3 +	12	0
2	5 / 8	4 / 12	9 / 13	12	−6
3	5 / −6	10 / + −8 (E)	10 / 6 −	6	8
D	6	15	9	Z = 162	
v_j	3	10	2		

4. New basic feasible solution with value Z = 162
5. Compute u_i, v_j and $c_{ij} - u_i - v_j$ and repeate ///

	1	2	3	s	u_i
1	3 / 6	10 / 8	2 / 6	12	0
2	5 / 0	4 / 12	9 / 5	12	2
3	5 / −6	10 / 3	10 / 3	6	8
D	6	15	9	Z = 138	
v_j	3	2	2		

6. New basic feasible sol

We have not reached optimality because entry (3,1) is negative

SOURCE \ DESTINATION	TODAY	TOMORROW	DAY AFTER	UNUSED	SUPPLY
TODAY	20	23	M	0	12
TOMORROW	M	40	43	0	4
DAY AFTER	M	M	45	0	8
DEMAND	6	7	8	3	

high cost because we cannot send cookies to the past nor store them for more than one day

unused amount of cookies produced (Σ supply $-$ Σ demand)

COIN FLIP

a. $\text{Prob}(HHTHH) = \frac{1}{32}$ (5 coin flips)

$\mathbb{E}[HHTHH \text{ nr. occurs}] = 32$

Head·start HH (longest sequence that ~~[illegible]~~ helps to get a new HHTHH)

$\mathbb{E}[HHTHH] = 32 + \mathbb{E}[HH]$

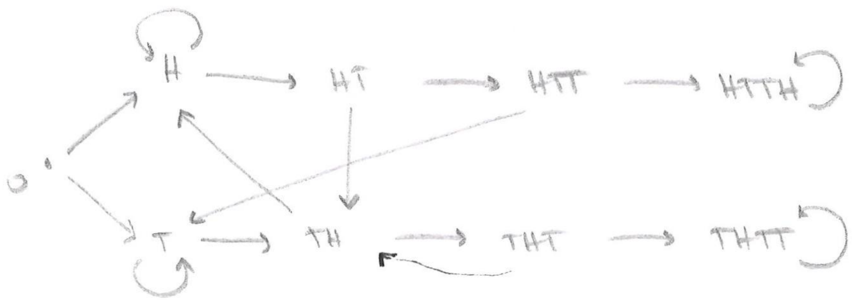
$\text{Prob}(HH \text{ nr. occurs}) = \frac{1}{4}$ $\quad\mathbb{E}[HH \text{ nr. occurs}] = 4$

Head·start H

$\mathbb{E}[H] = 2$

$\mathbb{E}[HHTHH] = 32 + 4 + 2 = 38$

b.



P_x = prob to reach HTTH before THTT starting from x

$$\begin{cases} P_0 = \frac{1}{2}P_H + \frac{1}{2}P_T \\ P_H = \frac{1}{2}P_H + \frac{1}{2}P_{HT} \\ P_{HT} = \frac{1}{2}P_{TH} + \frac{1}{2}P_{HTT} \\ P_{HTT} = \frac{1}{2}P_T + \frac{1}{2}P_{HTTH} \\ P_{HTTH} = 1 \\ P_T = \frac{1}{2}P_T + \frac{1}{2}P_{TH} \\ P_{TH} = \frac{1}{2}P_H + \frac{1}{2}P_{THT} \\ P_{THT} = \frac{1}{2}P_{TH} + \frac{1}{2}P_{THHT} \\ P_{THHT} = 0 \end{cases}$$

$P_{THHT} = 0$

$P_{HTTH} = 1$

$P_{THT} = \frac{1}{2} P_{TH}$

$P_{TH} = \frac{1}{2} P_H + \frac{1}{4} P_{TH}$ $\qquad$ $P_{TH} = \frac{2}{3} P_H$

$P_T = \frac{1}{2} P_T + \frac{1}{3} P_H$ $\qquad$ $P_T = \frac{2}{3} P_H$

$P_{HTT} = \frac{1}{3} P_H + \frac{1}{2}$

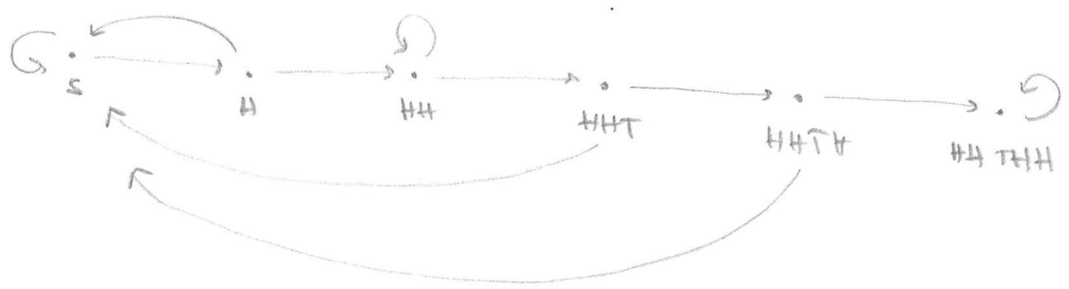
$P_{HT} = \frac{1}{2} P_H + \frac{1}{4}$

$P_H = \frac{1}{2} P_H + \frac{1}{2}\left(\frac{1}{2} P_H + \frac{1}{4}\right)$

$P_H = \frac{1}{2}$

$P_T = \frac{1}{3}$

$P_0 = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$



$$E_S = 1 + \frac{1}{2} E_H + \frac{1}{2} E_S$$

$$E_H = 1 + \frac{1}{2} E_{HH} + \frac{1}{2} E_S$$

$$E_{HH} = 1 + \frac{1}{2} E_{HH} + \frac{1}{2} E_{HHT}$$

$$E_{HHT} = 1 + \frac{1}{2} E_S + \frac{1}{2} E_{HHTH}$$

$$E_{HHTH} = 1 + \frac{1}{2} \cancel{E_{HHTHH}} + \frac{1}{2} E_S$$

$$E_{HHTHH} = 0$$

$$E_{HHTH} = 1 + \frac{1}{2} E_S$$

$$E_{HHT} = 1 + \frac{1}{2} E_S + \frac{1}{2}\left(1 + \frac{1}{2} E_S\right)$$

$$= 1 + \frac{1}{2} E_S + \frac{1}{2} + \frac{1}{4} E_S$$

$$= \frac{3}{2} + \frac{3}{4} E_S$$

$$E_{HH} = 1 + \frac{1}{2} E_{HH} + \frac{1}{2}\left(\frac{3}{2} + \frac{3}{4} E_S\right)$$

$$\frac{1}{2} E_{HH} = 1 + \frac{3}{4} + \frac{3}{8} E_S$$

$$= \frac{7}{4} + \frac{3}{8} E_S$$

$$E_{HH} = \frac{7}{2} + \frac{3}{4} E_S$$

$$E_H = 1 + \frac{1}{2}\left(\frac{7}{2} + \frac{3}{4} E_S\right) + \frac{1}{2} E_S$$

$$= 1 + \frac{7}{4} + \frac{3}{8} E_S + \frac{1}{2} E_S$$

$$= \frac{11}{4} + \cancel{\frac{3+4}{8}} \frac{7}{8} E_S$$

$$\frac{1}{16} E_S = \frac{19}{8}$$

$$E_S = 38$$

$$E_S = 1 + \frac{1}{2}\left(\frac{11}{4} + \frac{7}{8} E_S\right) + \frac{1}{2} E_S$$

$$= 1 + \frac{11}{8} + \frac{7}{16} E_S + \frac{1}{2} E_S$$

$$1 - \frac{7}{16} - \frac{1}{2} = \frac{16 - 7 - 8}{16} = \frac{1}{16}$$